Class Assignment

Question 1:

A small delivery company keeps track of packages waiting to be shipped.
Each package record contains:

- Package ID
- Destination city

Packages are stored in the order they arrive.

The system must support the following operations:

1. Add a new package at the end of the collection
2. Remove a package using its Package ID
3. Display all remaining packages in the order they were received

Once a package is removed, the surrounding packages must reconnect correctly so that the order remains intact.

Task

Write a C program that:

- Dynamically stores package records
- Allows deletion of any package using its ID
- Displays the updated collection after each operation

Note: Memory should be allocated and released dynamically.

Question 2:

A music player maintains a playlist where each song has:

- Song ID
- Song title

The user can:

1. Move forward to the next song
2. Move backward to the previous song
3. Insert a new song at any position in the playlist
4. Remove a song using its Song ID

Navigation must work correctly in both directions after every insertion or deletion.

Task

Write a C program that:

- Stores songs dynamically
- Supports forward and backward navigation
- Allows insertion and deletion at any position
- Displays the playlist from beginning to end and end to beginning

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Question 3:

In a multiplayer game lobby, players are arranged in a circle for turn-based gameplay.
Each player has:

- Player ID
- Player name

Rules:

1. Turns move in a fixed direction and never reach a “last” player
2. A player may leave the game at any time
3. New players can join the circle at any position

The turn system must continue smoothly even after players join or leave.

Task

Write a C program that:

- Maintains players in a continuous circular order
- Supports adding and removing players
- Simulates turn rotation for a given number of turns
- Displays all players starting from any chosen player